

4. Bud rate

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for generalized linear mixed-effects models (GLMM).

```

buds_df <-

  #' Load the raw data frame.
  df %>%

  #' Reshape the weekly bud counts from wide to long format.
  pivot_longer(cols = starts_with("buds_"),
               names_to = c(".value", "week"),
               names_sep = "_",) %>%

  #' Convert week and bud count to integers.
  mutate(week = as.integer(week),
         buds = as.integer(buds)) %>%

  #' Remove weeks with missing bud count.
  filter(!is.na(buds)) %>%

  #' Keep only the variables needed for the analysis.
  select(sleep_condition, lineage, tum_state, week, buds, id_unique, parent_unique,
         replicate_unique)

```

Set reference levels for the categorical variables.

```

buds_df$lineage <- relevel(factor(buds_df$lineage), ref = "HO_MT")
buds_df$sleep_condition <- relevel(factor(buds_df$sleep_condition), ref = "Control")
buds_df$tum_state <- relevel(factor(buds_df$tum_state), ref = "Healthy")

```

3 Model selection

3.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

buds_pois_R_00 <- glmmTMB(buds ~ week * lineage * sleep_condition, REML = TRUE, family =
poisson, data = buds_df)
buds_pois_R_01 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | id_unique), REML
= TRUE, family = poisson, data = buds_df)
buds_pois_R_02 <- glmmTMB(buds ~ week * lineage * sleep_condition + (week | id_unique),
REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_03 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | parent_unique),
REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_04 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_05 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | id_unique) + (1
| parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_06 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | id_unique) + (1
| replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_07 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | id_unique) + (1
| parent_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)

```

```

buds_pois_R_names <- paste0("buds_pois_R_", sprintf("%02d", c(0:7)))
buds_pois_R_objects <- mget(buds_pois_R_names)
print(head(do.call(model.sel, buds_pois_R_objects), 5))

```

```

## Model selection table
##
##          cond((Int)) dsp((Int)) cond(lng) cond(slp_cndt) cond(wek)
## buds_pois_R_04      0.3157      +      +      +      0.04045
## buds_pois_R_06      0.3157      +      +      +      0.04045
## buds_pois_R_07      0.3157      +      +      +      0.04045
## buds_pois_R_03      0.3192      +      +      +      0.04045
## buds_pois_R_01      0.3194      +      +      +      0.04045
##
##          cond(lng:slp_cndt) cond(lng:wek) cond(slp_cndt:wek)
## buds_pois_R_04      +      +      +
## buds_pois_R_06      +      +      +
## buds_pois_R_07      +      +      +
## buds_pois_R_03      +      +      +
## buds_pois_R_01      +      +      +
##
##          cond(lng:slp_cndt:wek)          random df      logLik      AICc
## buds_pois_R_04      +          c(r_u) 25 -6276.063 12602.4
## buds_pois_R_06      +          c(i_u)+c(r_u) 26 -6276.063 12604.4
## buds_pois_R_07      + c(i_u)+c(p_u)+c(r_u) 27 -6276.063 12606.4
## buds_pois_R_03      +          c(p_u) 25 -6287.509 12625.3
## buds_pois_R_01      +          c(i_u) 25 -6287.841 12626.0
##
##          delta weight
## buds_pois_R_04  0.00  0.668
## buds_pois_R_06  2.02  0.243
## buds_pois_R_07  4.04  0.088
## buds_pois_R_03 22.89  0.000
## buds_pois_R_01 23.56  0.000
## Models ranked by AICc(x)
## Random terms:
## c(r_u): cond(1 | replicate_unique)
## c(i_u): cond(1 | id_unique)
## c(p_u): cond(1 | parent_unique)

```

The model with the replicate id as a random effect (buds_pois_R_04) has the lowest AIC, this random effect structure is therefore selected.

3.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```

buds_pois_F_00 <- glmmTMB(buds ~ 1 + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_01 <- glmmTMB(buds ~ week + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_02 <- glmmTMB(buds ~ lineage + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_03 <- glmmTMB(buds ~ sleep_condition + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)

```

```

buds_pois_F_04 <- glmmTMB(buds ~ tum_state + (1 | replicate_unique), REML = FALSE, family
= poisson, data = buds_df)
buds_pois_F_05 <- glmmTMB(buds ~ week + lineage + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_06 <- glmmTMB(buds ~ week + sleep_condition + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_07 <- glmmTMB(buds ~ week + tum_state + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_08 <- glmmTMB(buds ~ lineage + sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_09 <- glmmTMB(buds ~ lineage + tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_10 <- glmmTMB(buds ~ sleep_condition + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_11 <- glmmTMB(buds ~ week * lineage + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_12 <- glmmTMB(buds ~ week * sleep_condition + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_13 <- glmmTMB(buds ~ week * tum_state + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_14 <- glmmTMB(buds ~ lineage * sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_15 <- glmmTMB(buds ~ lineage * tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_16 <- glmmTMB(buds ~ sleep_condition * tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_17 <- glmmTMB(buds ~ week + lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_18 <- glmmTMB(buds ~ week + lineage + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_19 <- glmmTMB(buds ~ week + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_20 <- glmmTMB(buds ~ lineage + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_21 <- glmmTMB(buds ~ week * lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_22 <- glmmTMB(buds ~ week * sleep_condition + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_23 <- glmmTMB(buds ~ week * tum_state + lineage + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_24 <- glmmTMB(buds ~ week * lineage + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_25 <- glmmTMB(buds ~ week * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_26 <- glmmTMB(buds ~ week * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_27 <- glmmTMB(buds ~ lineage * sleep_condition + week + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_28 <- glmmTMB(buds ~ lineage * tum_state + week + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_29 <- glmmTMB(buds ~ sleep_condition * tum_state + week + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_30 <- glmmTMB(buds ~ lineage * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

```

```

buds_pois_F_31 <- glmmTMB(buds ~ lineage * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_32 <- glmmTMB(buds ~ sleep_condition * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_33 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_34 <- glmmTMB(buds ~ week * lineage * tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_35 <- glmmTMB(buds ~ week * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_36 <- glmmTMB(buds ~ lineage * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

buds_pois_F_names <- paste0("buds_pois_F_", sprintf("%02d", c(0:36)))
buds_pois_F_objects <- mget(buds_pois_F_names)
print(head(do.call(model.sel, buds_pois_F_objects), 5))

```

```

## Model selection table
##               cond((Int)) dsp((Int)) cond(week) cond(lng) cond(slp_cndt)
## buds_pois_F_33      0.3061          +   0.04046          +          +
## buds_pois_F_21      0.3239          +   0.03678          +          +
## buds_pois_F_11      0.2886          +   0.03677          +          +
## buds_pois_F_24      0.3045          +   0.03678          +          +
## buds_pois_F_34      0.3268          +   0.03152          +          +
##               cond(tum_stt) cond(lng:week) cond(slp_cndt:week) cond(tum_stt:week)
## buds_pois_F_33                        +                      +
## buds_pois_F_21                        +
## buds_pois_F_11                        +
## buds_pois_F_24              +          +
## buds_pois_F_34              +          +
##               cond(lng:slp_cndt) cond(lng:tum_stt) cond(lng:slp_cndt:week)
## buds_pois_F_33              +                      +
## buds_pois_F_21
## buds_pois_F_11
## buds_pois_F_24
## buds_pois_F_34              +
##               cond(lng:tum_stt:week) df      logLik      AICc delta weight
## buds_pois_F_33                25 -6219.933 12490.1  0.00  0.906
## buds_pois_F_21                14 -6233.404 12494.9  4.76  0.084
## buds_pois_F_11                13 -6237.156 12500.4 10.25  0.005
## buds_pois_F_24                14 -6236.268 12500.6 10.48  0.005
## buds_pois_F_34                + 19 -6235.682 12509.5 19.38  0.000
## Models ranked by AICc(x)
## Random terms (all models):
##   cond(1 | replicate_unique)

```

The model with the interaction between the week, the lineage and the sleep condition has the lowest AIC and is therefore selected (buds_pois_F_33).

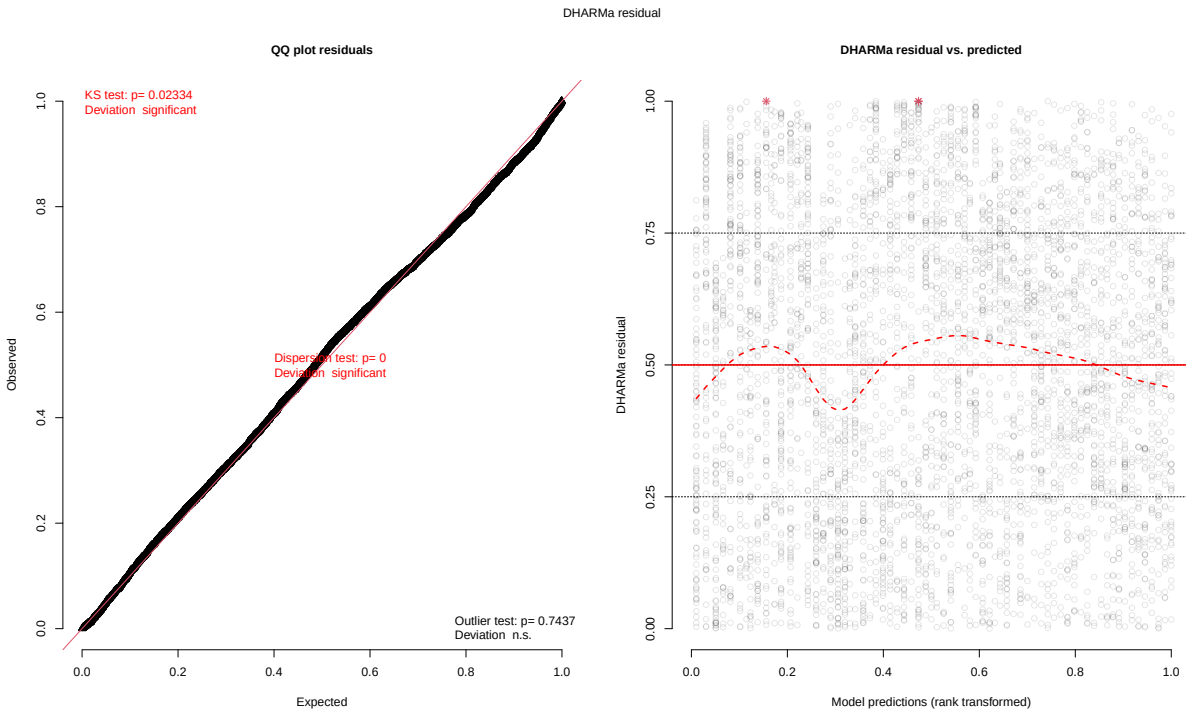
3.3 Goodness-of-fit evaluation

Refit the selected model.

```
buds_pois_mod_33 <- glmmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

Assess goodness-of-fit of the selected model using simulated residuals.

```
buds_pois_mod_33_res <- simulateResiduals(buds_pois_mod_33, plot = TRUE)
```



3.4 Selected model summary

Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_MT")
summary(glmmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
```

```

## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate Std. Error
## (Intercept)                        0.315678    0.094524
## week                               0.040455    0.023019
## lineageHC_MT                       -1.370099    0.202543
## lineageHO_SPC                      -0.609608    0.145661
## lineageHO_SPT                      -0.536080    0.145724
## lineageHO_VLN                     -1.637840    0.174026
## lineageHV_GAL                     -0.551752    0.140642
## sleep_conditionSleep deprivation  -0.036490    0.133237
## week:lineageHC_MT                   0.227553    0.044824
## week:lineageHO_SPC                   0.101016    0.034484
## week:lineageHO_SPT                   0.126876    0.034150
## week:lineageHO_VLN                   0.178343    0.040251
## week:lineageHV_GAL                   0.140067    0.033253
## week:sleep_conditionSleep deprivation -0.007714    0.033359
## lineageHC_MT:sleep_conditionSleep deprivation -0.184129    0.310125
## lineageHO_SPC:sleep_conditionSleep deprivation 0.369965    0.199130
## lineageHO_SPT:sleep_conditionSleep deprivation -0.117304    0.203784
## lineageHO_VLN:sleep_conditionSleep deprivation -0.239942    0.255964
## lineageHV_GAL:sleep_conditionSleep deprivation 0.281421    0.197316
## week:lineageHC_MT:sleep_conditionSleep deprivation -0.007118    0.069369
## week:lineageHO_SPC:sleep_conditionSleep deprivation -0.065524    0.048808
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.042134    0.048697
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.034530    0.059922
## week:lineageHV_GAL:sleep_conditionSleep deprivation -0.098026    0.049032
##
##                                     z value Pr(>|z|)
## (Intercept)                        3.340 0.000839 ***
## week                               1.757 0.078845 .
## lineageHC_MT                       -6.764 1.34e-11 ***
## lineageHO_SPC                      -4.185 2.85e-05 ***
## lineageHO_SPT                      -3.679 0.000234 ***
## lineageHO_VLN                      -9.411 < 2e-16 ***
## lineageHV_GAL                      -3.923 8.74e-05 ***
## sleep_conditionSleep deprivation  -0.274 0.784180
## week:lineageHC_MT                   5.077 3.84e-07 ***
## week:lineageHO_SPC                   2.929 0.003397 **
## week:lineageHO_SPT                   3.715 0.000203 ***
## week:lineageHO_VLN                   4.431 9.39e-06 ***
## week:lineageHV_GAL                   4.212 2.53e-05 ***
## week:sleep_conditionSleep deprivation -0.231 0.817126
## lineageHC_MT:sleep_conditionSleep deprivation -0.594 0.552697
## lineageHO_SPC:sleep_conditionSleep deprivation 1.858 0.063183 .
## lineageHO_SPT:sleep_conditionSleep deprivation -0.576 0.564865
## lineageHO_VLN:sleep_conditionSleep deprivation -0.937 0.348551
## lineageHV_GAL:sleep_conditionSleep deprivation 1.426 0.153797
## week:lineageHC_MT:sleep_conditionSleep deprivation -0.103 0.918272
## week:lineageHO_SPC:sleep_conditionSleep deprivation -1.343 0.179432
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.865 0.386910
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.576 0.564451

```

```
## week:lineageHV_GAL:sleep_conditionSleep deprivation -1.999 0.045582 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPC")
summary(glmmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML =
TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##              Estimate Std. Error z value
## (Intercept)    -0.29393    0.11084  -2.652
## week              0.14147    0.02568   5.510
## lineageHO_MT      0.60961    0.14566   4.185
## lineageHC_MT     -0.76049    0.21065  -3.610
## lineageHO_SPT      0.07353    0.15680   0.469
## lineageHO_VLN     -1.02823    0.18340  -5.607
## lineageHV_GAL      0.05786    0.15209   0.380
## sleep_conditionSleep deprivation  0.33347    0.14799   2.253
## week:lineageHO_MT -0.10102    0.03448  -2.929
## week:lineageHC_MT  0.12654    0.04624   2.736
## week:lineageHO_SPT  0.02586    0.03599   0.718
## week:lineageHO_VLN  0.07733    0.04183   1.849
## week:lineageHV_GAL  0.03905    0.03514   1.111
## week:sleep_conditionSleep deprivation -0.07324    0.03563  -2.056
## lineageHO_MT:sleep_conditionSleep deprivation -0.36996    0.19913  -1.858
## lineageHC_MT:sleep_conditionSleep deprivation -0.55409    0.31675  -1.749
## lineageHO_SPT:sleep_conditionSleep deprivation -0.48727    0.21372  -2.280
## lineageHO_VLN:sleep_conditionSleep deprivation -0.60991    0.26395  -2.311
## lineageHV_GAL:sleep_conditionSleep deprivation -0.08854    0.20757  -0.427
## week:lineageHO_MT:sleep_conditionSleep deprivation  0.06552    0.04881   1.343
## week:lineageHC_MT:sleep_conditionSleep deprivation  0.05841    0.07049   0.829
## week:lineageHO_SPT:sleep_conditionSleep deprivation  0.10766    0.05028   2.141
## week:lineageHO_VLN:sleep_conditionSleep deprivation  0.10005    0.06121   1.634
## week:lineageHV_GAL:sleep_conditionSleep deprivation -0.03250    0.05060  -0.642
##
##              Pr(>|z|)
## (Intercept)    0.008004 **
## week           3.59e-08 ***
## lineageHO_MT    2.85e-05 ***
```

```
## lineageHC_MT 0.000306 ***
## lineageHO_SPT 0.639123
## lineageHO_VLN 2.06e-08 ***
## lineageHV_GAL 0.703641
## sleep_conditionSleep deprivation 0.024236 *
## week:lineageHO_MT 0.003397 **
## week:lineageHC_MT 0.006213 **
## week:lineageHO_SPT 0.472475
## week:lineageHO_VLN 0.064496 .
## week:lineageHV_GAL 0.266492
## week:sleep_conditionSleep deprivation 0.039815 *
## lineageHO_MT:sleep_conditionSleep deprivation 0.063183 .
## lineageHC_MT:sleep_conditionSleep deprivation 0.080239 .
## lineageHO_SPT:sleep_conditionSleep deprivation 0.022611 *
## lineageHO_VLN:sleep_conditionSleep deprivation 0.020849 *
## lineageHV_GAL:sleep_conditionSleep deprivation 0.669683
## week:lineageHO_MT:sleep_conditionSleep deprivation 0.179432
## week:lineageHC_MT:sleep_conditionSleep deprivation 0.407332
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.032252 *
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.102159
## week:lineageHV_GAL:sleep_conditionSleep deprivation 0.520692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPT")
summary(glmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML =
TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##              Estimate Std. Error
## (Intercept)   -0.220402   0.110922
## week           0.167331   0.025225
## lineageHO_SPC  -0.073528   0.156800
## lineageHO_MT    0.536080   0.145724
## lineageHC_MT   -0.834019   0.210696
## lineageHO_VLN  -1.101760   0.183451
## lineageHV_GAL  -0.015672   0.152148
## sleep_conditionSleep deprivation -0.153794   0.154192
```

```

## week:lineageHO_SPC -0.025860 0.035994
## week:lineageHO_MT -0.126876 0.034150
## week:lineageHC_MT 0.100677 0.045995
## week:lineageHO_VLN 0.051467 0.041552
## week:lineageHV_GAL 0.013190 0.034816
## week:sleep_conditionSleep deprivation 0.034420 0.035476
## lineageHO_SPC:sleep_conditionSleep deprivation 0.487269 0.213719
## lineageHO_MT:sleep_conditionSleep deprivation 0.117304 0.203784
## lineageHC_MT:sleep_conditionSleep deprivation -0.066825 0.319706
## lineageHO_VLN:sleep_conditionSleep deprivation -0.122638 0.267480
## lineageHV_GAL:sleep_conditionSleep deprivation 0.398725 0.212035
## week:lineageHO_SPC:sleep_conditionSleep deprivation -0.107659 0.050278
## week:lineageHO_MT:sleep_conditionSleep deprivation -0.042134 0.048697
## week:lineageHC_MT:sleep_conditionSleep deprivation -0.049252 0.070412
## week:lineageHO_VLN:sleep_conditionSleep deprivation -0.007605 0.061126
## week:lineageHV_GAL:sleep_conditionSleep deprivation -0.140161 0.050497
## z value Pr(>|z|)
## (Intercept) -1.987 0.046923 *
## week 6.633 3.28e-11 ***
## lineageHO_SPC -0.469 0.639123
## lineageHO_MT 3.679 0.000234 ***
## lineageHC_MT -3.958 7.55e-05 ***
## lineageHO_VLN -6.006 1.90e-09 ***
## lineageHV_GAL -0.103 0.917960
## sleep_conditionSleep deprivation -0.997 0.318560
## week:lineageHO_SPC -0.718 0.472475
## week:lineageHO_MT -3.715 0.000203 ***
## week:lineageHC_MT 2.189 0.028607 *
## week:lineageHO_VLN 1.239 0.215487
## week:lineageHV_GAL 0.379 0.704790
## week:sleep_conditionSleep deprivation 0.970 0.331927
## lineageHO_SPC:sleep_conditionSleep deprivation 2.280 0.022611 *
## lineageHO_MT:sleep_conditionSleep deprivation 0.576 0.564865
## lineageHC_MT:sleep_conditionSleep deprivation -0.209 0.834433
## lineageHO_VLN:sleep_conditionSleep deprivation -0.458 0.646598
## lineageHV_GAL:sleep_conditionSleep deprivation 1.880 0.060044 .
## week:lineageHO_SPC:sleep_conditionSleep deprivation -2.141 0.032252 *
## week:lineageHO_MT:sleep_conditionSleep deprivation -0.865 0.386910
## week:lineageHC_MT:sleep_conditionSleep deprivation -0.699 0.484246
## week:lineageHO_VLN:sleep_conditionSleep deprivation -0.124 0.900992
## week:lineageHV_GAL:sleep_conditionSleep deprivation -2.776 0.005510 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_VLN")
summary(glmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML =
TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##

```

```

##          AIC          BIC      logLik -2*log(L)  df.resid
##    12602.1    12763.8    -6276.1    12552.1      4728
##
## Random effects:
##
## Conditional model:
##   Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##                                     Estimate Std. Error
## (Intercept)                       -1.322162    0.146120
## week                                0.218798    0.033019
## lineageHO_SPT                       1.101760    0.183451
## lineageHO_SPC                       1.028232    0.183400
## lineageHO_MT                        1.637840    0.174026
## lineageHC_MT                        0.267741    0.231171
## lineageHV_GAL                      1.086088    0.179438
## sleep_conditionSleep deprivation   -0.276432    0.218557
## week:lineageHO_SPT                 -0.051467    0.041552
## week:lineageHO_SPC                 -0.077327    0.041827
## week:lineageHO_MT                  -0.178343    0.040251
## week:lineageHC_MT                   0.049210    0.050688
## week:lineageHV_GAL                 -0.038276    0.040814
## week:sleep_conditionSleep deprivation 0.026816    0.049778
## lineageHO_SPT:sleep_conditionSleep deprivation 0.122638    0.267480
## lineageHO_SPC:sleep_conditionSleep deprivation 0.609907    0.263948
## lineageHO_MT:sleep_conditionSleep deprivation 0.239942    0.255964
## lineageHC_MT:sleep_conditionSleep deprivation 0.055813    0.355216
## lineageHV_GAL:sleep_conditionSleep deprivation 0.521363    0.262576
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.007605    0.061126
## week:lineageHO_SPC:sleep_conditionSleep deprivation -0.100054    0.061215
## week:lineageHO_MT:sleep_conditionSleep deprivation -0.034530    0.059922
## week:lineageHC_MT:sleep_conditionSleep deprivation -0.041648    0.078592
## week:lineageHV_GAL:sleep_conditionSleep deprivation -0.132556    0.061383
##                                     z value Pr(>|z|)
## (Intercept)                       -9.048 < 2e-16 ***
## week                                6.626 3.44e-11 ***
## lineageHO_SPT                       6.006 1.90e-09 ***
## lineageHO_SPC                       5.607 2.06e-08 ***
## lineageHO_MT                        9.411 < 2e-16 ***
## lineageHC_MT                        1.158 0.2468
## lineageHV_GAL                      6.053 1.42e-09 ***
## sleep_conditionSleep deprivation   -1.265 0.2059
## week:lineageHO_SPT                 -1.239 0.2155
## week:lineageHO_SPC                 -1.849 0.0645 .
## week:lineageHO_MT                  -4.431 9.39e-06 ***
## week:lineageHC_MT                   0.971 0.3316
## week:lineageHV_GAL                 -0.938 0.3483
## week:sleep_conditionSleep deprivation 0.539 0.5901
## lineageHO_SPT:sleep_conditionSleep deprivation 0.458 0.6466
## lineageHO_SPC:sleep_conditionSleep deprivation 2.311 0.0208 *
## lineageHO_MT:sleep_conditionSleep deprivation 0.937 0.3486

```

```
## lineageHC_MT:sleep_conditionSleep deprivation      0.157  0.8751
## lineageHV_GAL:sleep_conditionSleep deprivation      1.986  0.0471 *
## week:lineageHO_SPT:sleep_conditionSleep deprivation  0.124  0.9010
## week:lineageHO_SPC:sleep_conditionSleep deprivation -1.634  0.1022
## week:lineageHO_MT:sleep_conditionSleep deprivation  -0.576  0.5645
## week:lineageHC_MT:sleep_conditionSleep deprivation  -0.530  0.5962
## week:lineageHV_GAL:sleep_conditionSleep deprivation -2.159  0.0308 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HC_MT")
summary(glmmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML =
TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##              Estimate Std. Error
## (Intercept)   -1.054421   0.179134
## week              0.268008   0.038461
## lineageHO_VLN   -0.267741   0.231171
## lineageHO_SPT     0.834019   0.210696
## lineageHO_SPC     0.760491   0.210651
## lineageHO_MT      1.370099   0.202543
## lineageHV_GAL     0.818347   0.207211
## sleep_conditionSleep deprivation -0.220619   0.280053
## week:lineageHO_VLN -0.049210   0.050688
## week:lineageHO_SPT -0.100677   0.045995
## week:lineageHO_SPC -0.126537   0.046244
## week:lineageHO_MT  -0.227553   0.044824
## week:lineageHV_GAL -0.087487   0.045329
## week:sleep_conditionSleep deprivation -0.014832   0.060821
## lineageHO_VLN:sleep_conditionSleep deprivation -0.055813   0.355216
## lineageHO_SPT:sleep_conditionSleep deprivation  0.066825   0.319706
## lineageHO_SPC:sleep_conditionSleep deprivation  0.554094   0.316751
## lineageHO_MT:sleep_conditionSleep deprivation  0.184129   0.310125
## lineageHV_GAL:sleep_conditionSleep deprivation  0.465550   0.315600
## week:lineageHO_VLN:sleep_conditionSleep deprivation  0.041648   0.078592
## week:lineageHO_SPT:sleep_conditionSleep deprivation  0.049252   0.070412
## week:lineageHO_SPC:sleep_conditionSleep deprivation -0.058406   0.070488
```

```
## week:lineageHO_MT:sleep_conditionSleep deprivation 0.007118 0.069369
## week:lineageHV_GAL:sleep_conditionSleep deprivation -0.090908 0.070639
## z value Pr(>|z|)
## (Intercept) -5.886 3.95e-09 ***
## week 6.968 3.21e-12 ***
## lineageHO_VLN -1.158 0.246784
## lineageHO_SPT 3.958 7.55e-05 ***
## lineageHO_SPC 3.610 0.000306 ***
## lineageHO_MT 6.764 1.34e-11 ***
## lineageHV_GAL 3.949 7.84e-05 ***
## sleep_conditionSleep deprivation -0.788 0.430827
## week:lineageHO_VLN -0.971 0.331626
## week:lineageHO_SPT -2.189 0.028607 *
## week:lineageHO_SPC -2.736 0.006213 **
## week:lineageHO_MT -5.077 3.84e-07 ***
## week:lineageHV_GAL -1.930 0.053602 .
## week:sleep_conditionSleep deprivation -0.244 0.807336
## lineageHO_VLN:sleep_conditionSleep deprivation -0.157 0.875147
## lineageHO_SPT:sleep_conditionSleep deprivation 0.209 0.834433
## lineageHO_SPC:sleep_conditionSleep deprivation 1.749 0.080239 .
## lineageHO_MT:sleep_conditionSleep deprivation 0.594 0.552697
## lineageHV_GAL:sleep_conditionSleep deprivation 1.475 0.140179
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.530 0.596166
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.699 0.484246
## week:lineageHO_SPC:sleep_conditionSleep deprivation -0.829 0.407332
## week:lineageHO_MT:sleep_conditionSleep deprivation 0.103 0.918272
## week:lineageHV_GAL:sleep_conditionSleep deprivation -1.287 0.198115
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HV_GAL")
summary(glmmTMB(buds ~ week * lineage * sleep_condition + (1 | replicate_unique), REML =
TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## Estimate Std. Error z value
## (Intercept) -0.23607 0.10415 -2.267
## week 0.18052 0.02400 7.522
```

```

## lineageHC_MT -0.81835 0.20721 -3.949
## lineageHO_VLN -1.08609 0.17944 -6.053
## lineageHO_SPT 0.01567 0.15215 0.103
## lineageHO_SPC -0.05786 0.15209 -0.380
## lineageHO_MT 0.55175 0.14064 3.923
## sleep_conditionSleep deprivation 0.24493 0.14554 1.683
## week:lineageHC_MT 0.08749 0.04533 1.930
## week:lineageHO_VLN 0.03828 0.04081 0.938
## week:lineageHO_SPT -0.01319 0.03482 -0.379
## week:lineageHO_SPC -0.03905 0.03514 -1.111
## week:lineageHO_MT -0.14007 0.03325 -4.212
## week:sleep_conditionSleep deprivation -0.10574 0.03593 -2.943
## lineageHC_MT:sleep_conditionSleep deprivation -0.46555 0.31560 -1.475
## lineageHO_VLN:sleep_conditionSleep deprivation -0.52136 0.26258 -1.986
## lineageHO_SPT:sleep_conditionSleep deprivation -0.39873 0.21203 -1.880
## lineageHO_SPC:sleep_conditionSleep deprivation 0.08854 0.20757 0.427
## lineageHO_MT:sleep_conditionSleep deprivation -0.28142 0.19732 -1.426
## week:lineageHC_MT:sleep_conditionSleep deprivation 0.09091 0.07064 1.287
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.13256 0.06138 2.159
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.14016 0.05050 2.776
## week:lineageHO_SPC:sleep_conditionSleep deprivation 0.03250 0.05060 0.642
## week:lineageHO_MT:sleep_conditionSleep deprivation 0.09803 0.04903 1.999
## Pr(>|z|)
## (Intercept) 0.02341 *
## week 5.38e-14 ***
## lineageHC_MT 7.84e-05 ***
## lineageHO_VLN 1.42e-09 ***
## lineageHO_SPT 0.91796
## lineageHO_SPC 0.70364
## lineageHO_MT 8.74e-05 ***
## sleep_conditionSleep deprivation 0.09240 .
## week:lineageHC_MT 0.05360 .
## week:lineageHO_VLN 0.34833
## week:lineageHO_SPT 0.70479
## week:lineageHO_SPC 0.26649
## week:lineageHO_MT 2.53e-05 ***
## week:sleep_conditionSleep deprivation 0.00325 **
## lineageHC_MT:sleep_conditionSleep deprivation 0.14018
## lineageHO_VLN:sleep_conditionSleep deprivation 0.04708 *
## lineageHO_SPT:sleep_conditionSleep deprivation 0.06004 .
## lineageHO_SPC:sleep_conditionSleep deprivation 0.66968
## lineageHO_MT:sleep_conditionSleep deprivation 0.15380
## week:lineageHC_MT:sleep_conditionSleep deprivation 0.19812
## week:lineageHO_VLN:sleep_conditionSleep deprivation 0.03081 *
## week:lineageHO_SPT:sleep_conditionSleep deprivation 0.00551 **
## week:lineageHO_SPC:sleep_conditionSleep deprivation 0.52069
## week:lineageHO_MT:sleep_conditionSleep deprivation 0.04558 *
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

4 Estimated marginal means

4.1 Table

Compute the estimated marginal means of weekly bud production for every combination of week, sleep condition, and lineage, based on the selected model.

```
buds_emmeans <-  
  as.data.frame(ggemmeans(buds_pois_mod_33,  
    terms = c("week [all]", "sleep_condition", "lineage"),  
    bias_correction = TRUE)) %>%  
  arrange(facet, group, x)
```

4.2 Plots

4.3 Common functions

Function to plot, for a single lineage, the predicted weekly bud production by sleep condition.

```
buds_plot_treatment <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,  
  show_y_title = FALSE, show_x_title = FALSE) {  
  data_sub <- subset(buds_emmeans, facet == lineage)  
  
  ggplot(data_sub, aes(x = x, y = predicted, color = group, fill = group)) +  
    geom_line(linewidth = 0.8) +  
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +  
    labs(x = if(show_x_title) "Age (weeks)" else NULL,  
      y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"  
      else NULL,  
      title = NULL) +  
    theme_bw() +  
    theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else  
      element_blank(),  
      axis.ticks.x = if(show_x_text) element_line() else element_blank(),  
      axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else  
      element_blank(),  
      axis.ticks.y = if(show_y_text) element_line() else element_blank(),  
      axis.title.x = element_text(color = "black", size = 14, margin = margin(t =  
        10)),  
      axis.title.y.right = element_text(color = "black", size = 14, angle = 90,  
        margin = margin(l = 25)),  
      axis.line.y.right = element_blank(),  
      plot.margin = margin(l = 5, r = 5, t = 5, b = 5),  
      legend.position = "none") +  
    scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +  
    scale_y_continuous(position = "right") +  
    coord_cartesian(ylim = c(0, 3.5)) +  
    scale_color_manual(values = sleep_condition_colors) +  
    scale_fill_manual(values = sleep_condition_colors) +  
    annotate("text", x = 1.2, y = 3.3, label = lineage,  
      color = "black", fontface = "bold", size = 5, hjust = 0, vjust = 1)}
```

Function to plot, for a single lineage, the predicted weekly bud production pooled across sleep conditions

```
buds_plot_lineage <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)
  data_lineage <- aggregate(cbind(predicted, conf.low, conf.high) ~ x, data = data_sub,
FUN = mean)
  data_lineage <- data_lineage[order(data_lineage$x), ]

  ggplot(data_lineage, aes(x = x, y = predicted)) +
    geom_line(linewidth = 0.8, color = "black") +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, fill = "black",
linewidth = 0.3) +
    labs(x = if(show_x_title) "Age (weeks)" else NULL,
         y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
         else NULL,
         title = NULL) +
    theme_bw() +
    theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.x = if(show_x_text) element_line() else element_blank(),
          axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
element_blank(),
          axis.ticks.y = if(show_y_text) element_line() else element_blank(),
          axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
10)),
          axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
margin = margin(l = 25)),
          axis.line.y.right = element_blank(),
          plot.margin = margin(l = 5, r = 5, t = 5, b = 5)) +
    scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
    scale_y_continuous(position = "right") +
    coord_cartesian(ylim = c(0, 3.5)) +
    annotate("text", x = 1.2, y = 3.3, label = lineage,
            color = "black", fontface = "bold", size = 5, hjust = 0, vjust = 1) }
```

4.4 Panel A: interaction between sleep condition and lineage

```
A1 <- buds_plot_treatment("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
A2 <- buds_plot_treatment("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
A3 <- buds_plot_treatment("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
A4 <- buds_plot_treatment("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
A5 <- buds_plot_treatment("HC_MT", show_y_text = FALSE, show_x_text = TRUE)
A6 <- buds_plot_treatment("HV_GAL", show_y_text = TRUE, show_x_text = TRUE,
                        show_x_title = TRUE, show_y_title = TRUE)
```

Combine A1 to A6 into panel “A”

```
A <- (A1 + A2 + A3) / (A4 + A5 + A6) + plot_layout(heights = c(1, 1))
```

4.5 Panel B: lineage effect

```
B1 <- buds_plot_lineage("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
B2 <- buds_plot_lineage("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
B3 <- buds_plot_lineage("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
B4 <- buds_plot_lineage("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
B5 <- buds_plot_lineage("HC_MT", show_y_text = FALSE, show_x_text = TRUE)
B6 <- buds_plot_lineage("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_x_title =
TRUE, show_y_title = TRUE)
```

Combine B1 to B6 into panel “B”

```
B <- (B1 + B2 + B3) / (B4 + B5 + B6) + plot_layout(heights = c(1, 1))
```

Extract legend from the plot A6 to use later on the full figure.

```
legend_treatment <- cowplot::get_legend(A6 +
  labs(color = NULL, fill = NULL) +
  theme(legend.position = "right", legend.text =
    element_text(size = 13)))
```

4.6 Assembly

Add a letter tag (A/B) to the first sub-plot of each panel.

```
A[[1]][[1]] <- A[[1]][[1]] + labs(tag = "A") +
  theme(plot.tag = element_text(size = 20, face = "bold"))
B[[1]][[1]] <- B[[1]][[1]] + labs(tag = "B") +
  theme(plot.tag = element_text(size = 20, face = "bold"))
```

Convert each panel into a graphical object.

```
A <- patchwork::patchworkGrob(A)
B <- patchwork::patchworkGrob(B)
```

Build standalone title graphical objects for each panel.

```
interaction_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Interaction between sleep condition and lineage",
    fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

lineage_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Lineage effect",
    fontface = "bold", size = 16, x = 0.5, hjust = 0.5)
```

Stack the two titles and their corresponding panels vertically.

```

left_column <- cowplot::plot_grid(interaction_title_grob,
                                   A,
                                   lineage_title_grob,
                                   B,
                                   ncol = 1,
                                   align = "v",
                                   axis = "x",
                                   rel_heights = c(0.05, 1, 0.05, 1))

```

Shift the sleep-condition legend to the right, to align it with the plots.

```

legend_treatment_shifted <- cowplot::plot_grid(NULL, legend_treatment, ncol = 2,
rel_widths = c(0.6, 0.1))

```

Place the legend in a right-hand column.

```

right_column <- cowplot::plot_grid(legend_treatment_shifted, NULL, ncol = 1, rel_heights
= c(0.9, 4))

```

Combine the panels and the legend.

```

final_buds_panel <- cowplot::plot_grid(left_column, right_column, ncol = 2, rel_widths =
c(1, 0.16)) +
  theme(plot.margin = margin(l = 5, r = 80, t = 5, b = 5))

```

Show the final figure.

```

print(final_buds_panel)

```

